Unit 4: Covariance Matrices, PCA, and Stochastic Calculus

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Covariance Matrices

Covariance Matrix

Definition

Suppose $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^T$ is a multivariate random variable, and $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)^T$. The **covariance matrix** of $\boldsymbol{X}$ is

$$\Sigma = E\left[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T\right].$$

Notice that

$$\Sigma_{ij} = \begin{cases} \text{Var}(X_i), & i = j \\ \text{Cov}(X_i, X_j), & i \neq j. \end{cases}$$

Multivariate Normal Distribution

Definition

The **multivariate normal distribution** or **Gaußian distribution** of dimension k has probability density function

$$f(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right),$$

where $\boldsymbol{\mu}$ is in $\mathbb{R}^k$ and Σ is the distribution's $k \times k$ covariance matrix. To denote that $\boldsymbol{X}$ follows a multivariate normal distribution, we write $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ or $\boldsymbol{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$.

Bivariate Normal Distribution Python Example

Example

Sample 100 points from $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where $\boldsymbol{\mu} = (0,0)^T$ and $\Sigma =$

(a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix}$

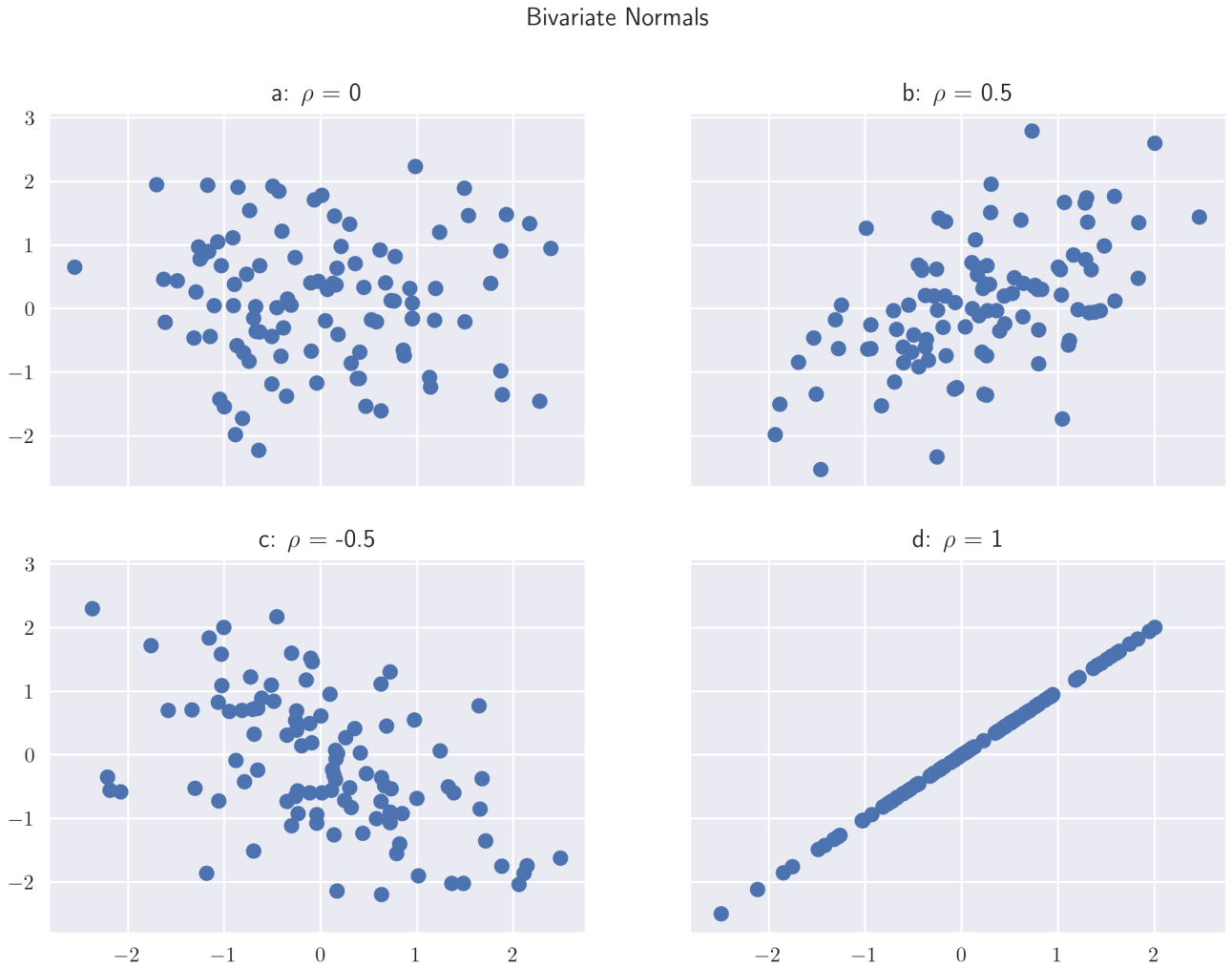
(d) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Graph the results.

Bivariate Normal Distribution Example

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import multivariate_normal

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams[
    'figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# Create list for problem parts
parts = ['a', 'b', 'c', 'd']

# Create list of covariance matrices
covs = [np.array([[1, 0], [0, 1]]),
        np.array([[1, 0.5], [0.5, 1]]),
        np.array([[1, -0.5], [-0.5, 1]]),
        np.array([[1, 1], [1, 1]])]

# Set up subplots
fig, ax = plt.subplots(2, 2, sharex = True,
    sharey = True, figsize = (10, 7))

# Loop over titles and covariance matrices
for i, part, cov in zip(range(4), parts,
    covs):

    # Get the row and column
    row, col = i // 2, i %2

    # Generate values
    vals = multivariate_normal.rvs(mean =
      np.zeros(2), cov = cov, size = 100)

    # Get x- and y-coordinates
    x, y = zip(*vals)

    # Plot the values
    ax[row, col].scatter(x, y)

    # Get title
    title = part + r': $\rho$ = ' + str(cov
      [0, 1])

    # Give the plot a title
    ax[row, col].title.set_text(title)

# Give entire figure title
fig.suptitle('Bivariate Normals')

# Save the figure
plt.savefig(path + r'ex4-1.png')

plt.show()
```

Bivariate Normal Distribution Example Result



Sample Covariance

In the sample covariance matrix, we divide by $n-1$ instead of n. Using the `pandas` data frame `df` the sample covariance matrix would be `df.cov()`.

Sample Covariance Matrix Example

Here is some code to get the sample covariance matrix for the *current* S&P constituents. The uploaded data are the daily constituents' returns from January 1, 2020 to April 30, 2025 which are available over the whole time range. The data source is *Yahoo! Finance*. Check out the Unit 4 Code Snippets to see how the data were extracted.

```
# Import modules
import pandas as pd

# Load in data; make date the index
data = pd.read_csv(data_path, index_col = 'Date')

# Get covariance matrix
S = data.cov()

S
```

Sample Covariance Matrix Result

The output looks like this:

	A	AAPL	ABBV	ABT	ACGL	ACN	ADBE	ADI	ADM	ADP	...
A	0.000358	0.000202	0.000109	0.000172	0.000156	0.000202	0.000229	0.000254	0.000133	0.000171	...
AAPL	0.000202	0.000427	0.000100	0.000152	0.000161	0.000227	0.000311	0.000297	0.000130	0.000195	...
ABBV	0.000109	0.000100	0.000250	0.000116	0.000124	0.000107	0.000100	0.000110	0.000093	0.000117	...
ABT	0.000172	0.000152	0.000116	0.000270	0.000136	0.000157	0.000167	0.000162	0.000110	0.000155	...
ACGL	0.000156	0.000161	0.000124	0.000136	0.000449	0.000194	0.000148	0.000198	0.000201	0.000215	...
...	...	...	...	...	...	...	...	...	...	...	...
XYL	0.000211	0.000205	0.000099	0.000154	0.000228	0.000213	0.000200	0.000267	0.000173	0.000208	...
YUM	0.000132	0.000148	0.000090	0.000115	0.000174	0.000153	0.000133	0.000175	0.000129	0.000152	...
ZBH	0.000155	0.000146	0.000111	0.000140	0.000195	0.000173	0.000139	0.000193	0.000150	0.000156	...
ZBRA	0.000280	0.000299	0.000103	0.000173	0.000205	0.000272	0.000313	0.000374	0.000175	0.000224	...
ZTS	0.000207	0.000205	0.000118	0.000171	0.000154	0.000193	0.000216	0.000209	0.000115	0.000177	...

489 rows × 489 columns

Note: There are fewer than 500 columns because some of the current S&P constituents weren't publicly traded companies in 2020.

Principal Component Analysis (PCA)

Convention

Let's assume that

$$i < j \qquad \text{implies} \qquad \lambda_i \geq \lambda_j.$$

In other words, we've rearrange our eigenvalues and corresponding eigenvectors so that the eigenvalues are in descending order.

Eigenvectors and Eigenvalues

From the spectral theorem, we know that Σ is diagonalizable, since it is symmetric. If Σ is $k \times k$, and the respective eigenvectors and eigenvalues are $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k$ and $\lambda_1, \lambda_2, \ldots, \lambda_k$. Then the total variance is $\lambda_1 + \lambda_2 + \ldots + \lambda_k$, and the fraction of variance explained by eigenvector $\boldsymbol{v}_i$ is

$$\frac{\lambda_i}{\lambda_1 + \lambda_2 + \ldots + \lambda_i + \ldots + \lambda_k}.$$

Eigenvectors and Eigenvalues Example

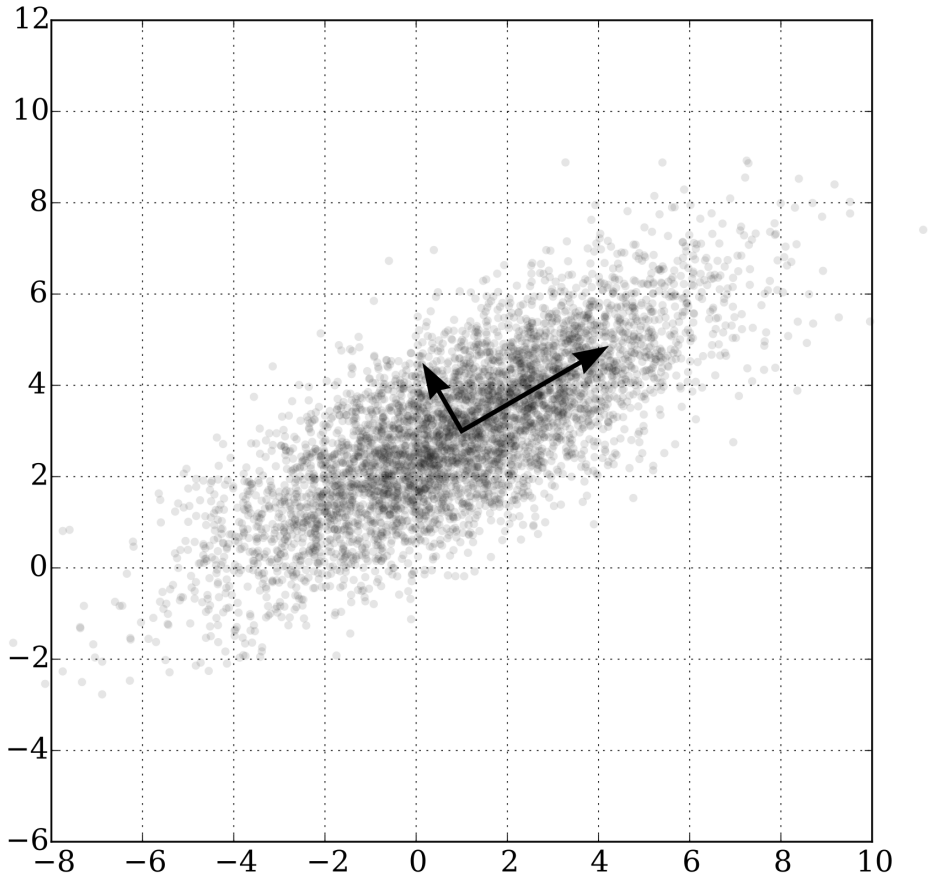
Example

Consider $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$. Find the eigenvectors and eigenvalues.

Principal Component Analysis

Principal component analysis (PCA) is a dimension reduction technique. The data are linearly transformed into a new coordinate system of eigenvectors (principal components) corresponding to the largest eigenvalues.

Principal Component Analysis Figure



Principal Component Analysis Steps

1. Collect your data.
2. Standardize or demean your data.
3. Determine how much of the variance you need to explain or how many components are usable for your project.
4. Get the orthonormal basis of eigenvectors and eigenvalues.
5. Subset the orthonormal basis to the vectors corresponding to the largest ℓ eigenvalues, where the value of ℓ is based on step 3.
6. Calculate the "loadings" of each observation on the remaining basis elements.

Orthonormal Basis

Suppose Σ is the covariance matrix corresponding to the $n \times k$ matrix X.

The eigenvectors of Σ form the orthonormal eigenbasis $(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k)$.

So, for any $\boldsymbol{u}$ in $\mathbb{R}^k$, we can write

$$\boldsymbol{u} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \ldots + \alpha_\ell \boldsymbol{v}_\ell + \ldots + \alpha_k \boldsymbol{v}_k$$

for unique α_i

What are Loadings?

Because the basis $(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k)$ is orthognal, we know

$$\boldsymbol{u} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \ldots + \alpha_\ell \boldsymbol{v}_\ell + \ldots + \alpha_k \boldsymbol{v}_k$$

implies the **loading**

$$\alpha_i = \frac{\boldsymbol{u} \bullet \boldsymbol{v}_i}{\|\boldsymbol{v}_i\|^2} = \boldsymbol{u} \bullet \boldsymbol{v}_i.$$

The First ℓ Principapal Components

If we have

$$\boldsymbol{u} = \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \ldots + \alpha_\ell \boldsymbol{v}_\ell + \ldots + \alpha_k \boldsymbol{v}_k$$

then the best way to reduce linearly the dimension while preserving as much of the explained variance as possible is to assume

$$\boldsymbol{u} \approx \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \ldots + \alpha_\ell \boldsymbol{v}_\ell.$$

In terms of a basis of the first ℓ principal components, we're saying

$$\boldsymbol{u} \approx \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_\ell \end{pmatrix}.$$

Loadings for Your Data

Suppose X is the $n \times k$ matrix from before whose covariance matrix Σ was used to find an orthonormal basis of eigenvectors. Furthrmore, suppose x_i is the i-th row of X, and $\boldsymbol{\mu}$ is the vector whose j-th row is the mean of the j-th column of X. Then the p-th loadings corresponding to $\boldsymbol{x}_i$ are

$$(\boldsymbol{x}_i - \boldsymbol{\mu}^T)^T \bullet \boldsymbol{v}_p = (\boldsymbol{x}_i - \boldsymbol{\mu}^T)\boldsymbol{v}_p.$$

The matrix of loadings for the first ℓ principal components is

$$(X - \boldsymbol{\mu}^T)\left(\boldsymbol{v}_1 \ \boldsymbol{v}_2 \ \ldots \boldsymbol{v}_\ell\right),$$

where $X - \boldsymbol{\mu}^T$ means subtraction of the $\boldsymbol{\mu}^T$ from each row of X.

Principal Component Analysis Example

Example

Generate 100 points from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where

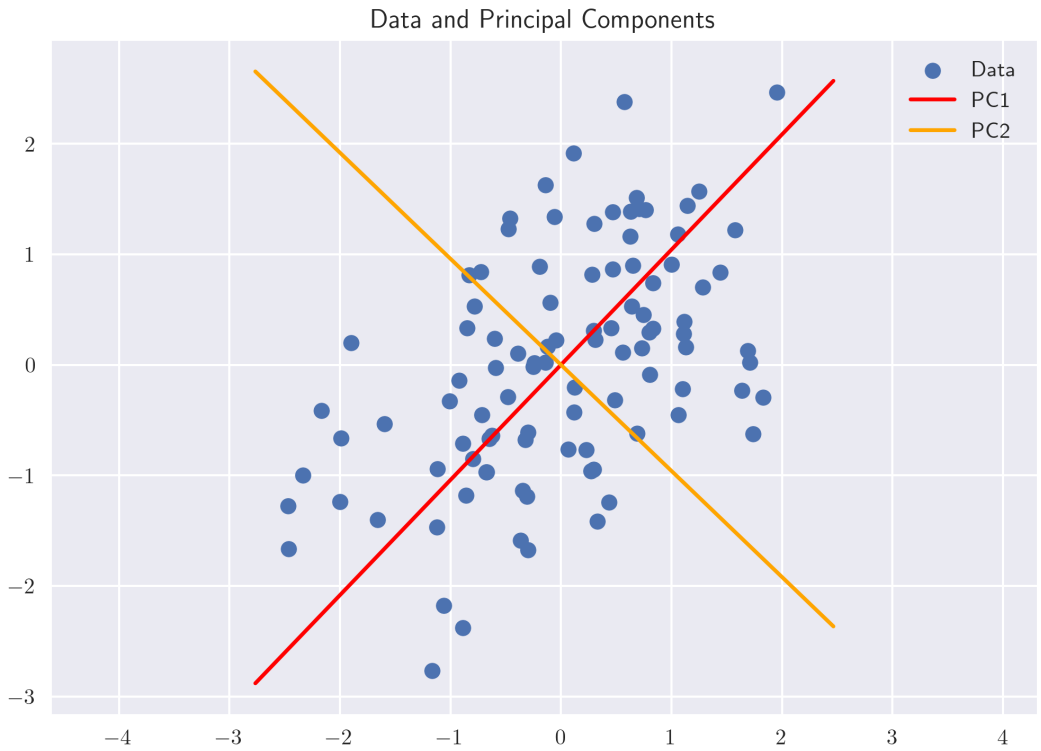
$$\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}.$$

Then plot lines in the directions of the first two principal components.

Principal Component Analysis Example

```python
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import multivariate_normal

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set random seed
np.random.seed(0)

# Get covariance matrix
cov = np.array([[1, 0.5], [0.5, 1]])

# Generate data
X = multivariate_normal.rvs(
    mean = np.zeros(2),
    cov = cov,
    size = 100)

# Demean data, though mean already close to
    zero
X -= X.mean(axis = 0)

# Calculate covariance matrix
S = np.cov(X, rowvar = False, ddof = 1)

# Get eigenvalues and eigenvectors
evals, evecs = np.linalg.eigh(S)

# Index to sort eigenvalues into descending
    order
idx = evals.argsort()[::-1]

# Sort eigenvalues and -vectors
evals, evecs = evals[idx], evecs[:, idx]

# Get slopes
m_1 = evecs[1, 0]/evecs[0, 0]
m_2 = evecs[1, 1]/evecs[0, 1]

# Get x-values for plot
x_vals = np.linspace(X.min(), X.max())

# Plot graph
plt.scatter(*X.T, label = 'Data')
plt.plot(x_vals, m_1 * x_vals,
    color = 'red', label = 'PC1')
plt.plot(x_vals, m_2 * x_vals,
    color = 'orange', label = 'PC2')

# Add legend
plt.legend()

# Make aspect ratio the same, so PC's look
    perp
plt.axis('equal')

# Add title
plt.title('Data and Principal Components')

# Save the figure
plt.savefig(path + r'ex4-2.png')

plt.show()
```

Principal Component Analysis Example Result



Principal Component Analysis Example

Example

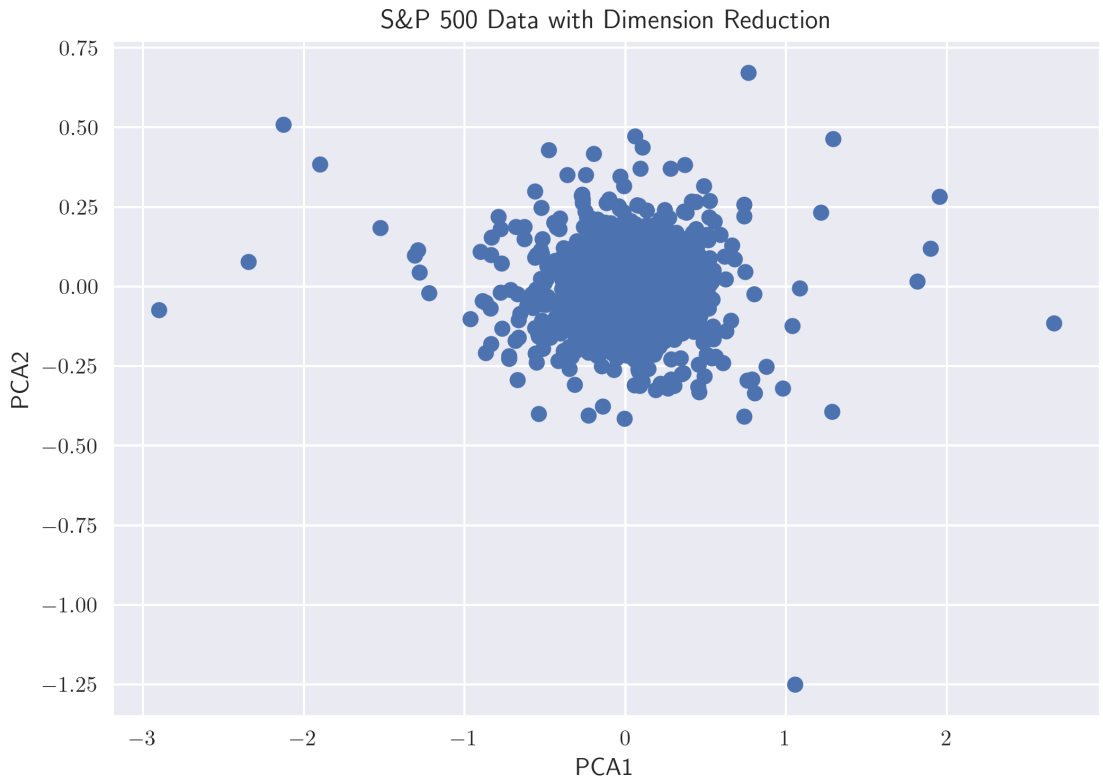
Use PCA to represent the S&P 500 constituent data from before on a two dimensional scatter plot.

Principal Component Analysis Example

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt

# Increase resolution
plt.rcParams['figure.dpi'] = 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Get S&P 500 return cov matrix
S = data.cov()

# Get the eigenvalues and eigenvectors
evals, evecs = np.linalg.eigh(S)

# Index to sort eigenvalues into descending
      order
idx = evals.argsort()[::-1]

# Change order
evals, evecs = evals[idx], evecs[:, idx]

# Convert the observations to a numpy array
X = data.values
```

```
# Demean X
X -= X.mean(axis = 0)

# Calculate loadings using the dot product
loadings = X @ evecs[:, 0:2]

# Unpack results
x, y = zip(*loadings)

# Get scatter plot
plt.scatter(x, y)

# Create x- and y-labels
plt.xlabel('PCA1')
plt.ylabel('PCA2')

# Give the plot a title
plt.title(r'S\&P 500 Data with Dimension
      Reduction')

# Save the figure
plt.savefig(path + r'ex4-3.png')

plt.show()
```

Principal Component Analysis Example Result



Reconstruction

If we have data with mean $\boldsymbol{\mu}$, and we used the loadings of the first ℓ eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_\ell$ to approximate row vector $\boldsymbol{x}$, then this is making the assumption

$$\boldsymbol{x}^T \approx \boldsymbol{\mu} + \alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \ldots + \alpha \boldsymbol{v}_\ell,$$

where α_i is the loading corresponding to $\boldsymbol{v}_i$.

Why Standardize?

PCA results are affected by the scale of your data. However, in many applications scale is already known and users are more interested in correlations within the data.

Clipping Covariance Matrices

Positive Definite Matrices

Using the dot product, a matrix S is *positive definite* if

$$\boldsymbol{x}^T S\boldsymbol{x} > 0 \qquad \text{for all} \qquad \boldsymbol{x} \neq \mathbf{0}.$$

Since a covariance matrix is diagonalizable due to the spectral theorem, a covariance matrix will be positive definite if and only if all its eigenvalues are positive.

Negative and Zero Eigenvalues

- When S is a covariance matrix, it never makes sense to have negative eigenvalues. These results are simply numerical errors.
- A zero eigenvalue indicates one variable can be written as a linear combination of the others, which may or may not make sense given the context.

Negative and Zero Eigenvalues Example

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Set the random seed
np.random.seed(0)

# Generate normal random variables
X = norm.rvs(size = (50, 100))

# Get the covariance matrix
S = np.cov(X, rowvar = False)

# Get the eigenvalues
evals, _ =  np.linalg.eigh(S)

print(f'The sample covariance matrix has {np.sum(np.isclose(evals, 0))}',
      r'eigenvalues numerically indistinguishable from 0.')
```

It says there are 51 eigenvalues numerically indistinguishable from 0. The true covariance matrix is the identity I and therefore all the true eigenvalues equal 1.

Clip Covariance Matrices

One solution is to "clip" the sample covariance matrix. Simply replace all the eigenvalues that are too small with something bigger and reconstruct the covariance matrix.

Clip Covariance Matrices Example

Using the results from before.

```
# Get the standard deviations
stds = np.sqrt(np.diag(S))

# Get the correlation matrix
C = np.diag(1/stds) @ S @ np.diag(1/stds)

# Get the eigenvalues and vectors of C
evals_c, evecs_c = np.linalg.eigh(C)

# Make lam_min small positive outside of np
    .isclose threshold
lam_min = 1e-7

# Replace eigenvalues that are too small
evals_c[evals_c < lam_min] = lam_min

# Reconstruct correlation matrix
C_new = evecs_c @ np.diag(evals_c) @
    evecs_c.T
```

```
# Make sure still correlation matrix
C_new = (np.diag(np.sqrt(1/np.diag(C_new)))
     @ C_new
          @ np.diag(np.sqrt(1/np.diag(
    C_new))))

# Multiply by standard deviations to make
    it covariance matrix
S_new = np.diag(stds) @ C_new @ np.diag(
    stds)

# Get the eigenvalues
evals_new, _ =  np.linalg.eigh(S_new)

print(f'The new covariance matrix has {np.
    sum(np.isclose(evals_new, 0))}',
     r'eigenvalues numerically
    indistinguishable from 0.')
```

Now it says all the eigenvalues are positive! Inspection shows that the covariance matrix estimate is otherwise very similar to the sample covariance matrix.

Marchenko–Pastur Theorem

Theorem

Consider a matrix of independent and identically distributed random observations X of size $T \times N$, where the underlying process generating the observations has mean 0 and variance σ^2. If $q = T/N > 1$ is constant, the covariance matrix $S = \frac{1}{T} X^T X$ has eigenvalues that converges to a distribution with probability density function

$$f(\lambda) = \begin{cases} \frac{q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda}, & \lambda_- \leq \lambda \leq \lambda_+ \\ 0, & \text{otherwise} \end{cases}$$

where

$$\lambda_- = \sigma^2 \left(1 - \sqrt{\frac{1}{q}}\right)^2 \quad \text{and} \quad \lambda_+ = \sigma^2 \left(1 + \sqrt{\frac{1}{q}}\right)^2.$$

Marchenko–Pastur Example

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm
import time

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set the random seed
np.random.seed(0)

# Start the clock
start_time = time.perf_counter()

# Generate normal random variables
X = norm.rvs(size = (100_000, 10_000))
```

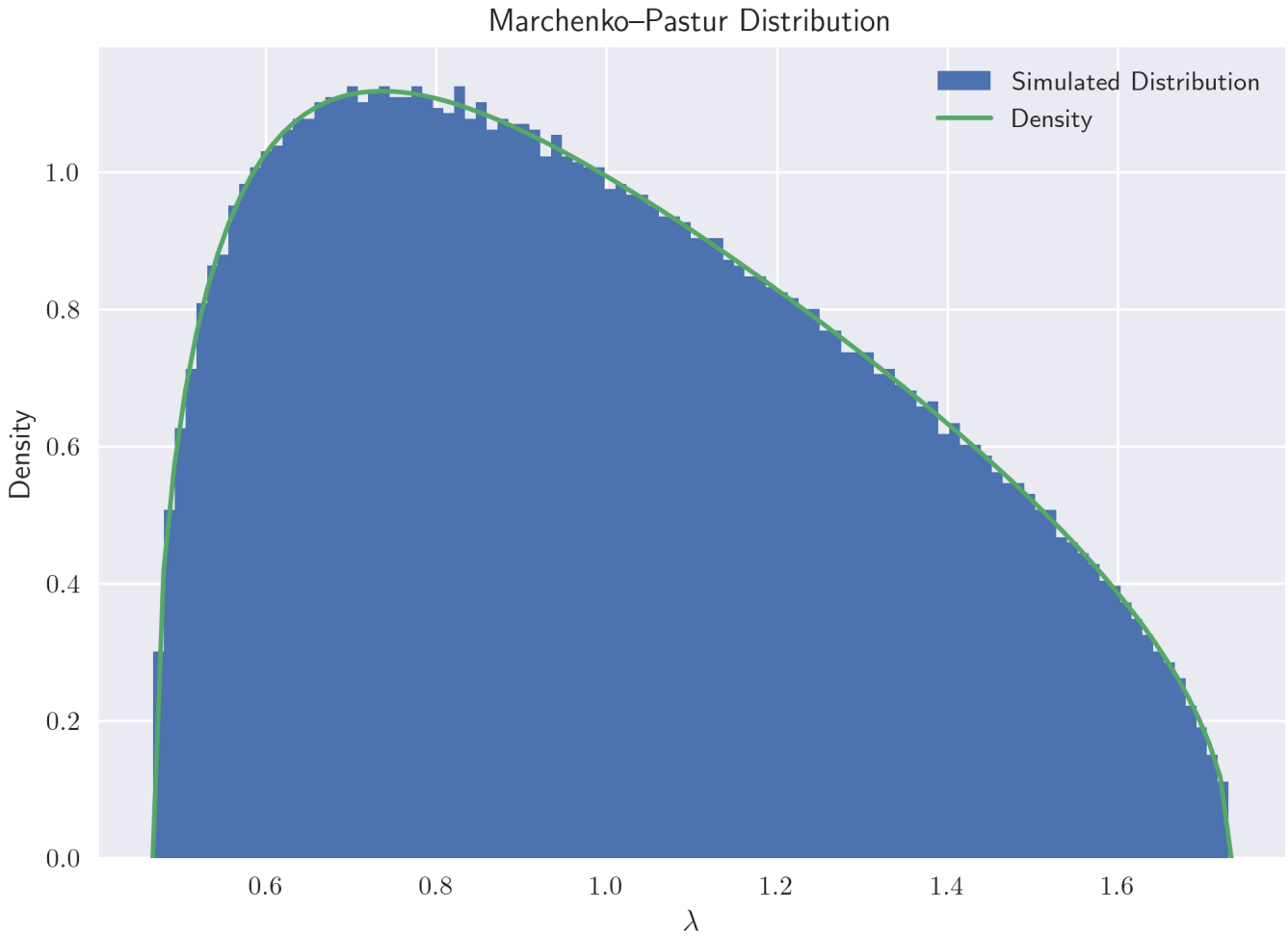
Marchenko–Pastur Example

```
# Get the number of observations and variables
T, N = X.shape

# Get correlation matrix
C = np.corrcoef(X, rowvar = False)

# q is the number of observations divided by the number of variables
q = T/N

# Get eigenvalues
evals, _ = np.linalg.eigh(C)

# Get support of Marchenko-Pastur distribution
lam_minus, lam_plus = (1 - np.sqrt(1/q))**2, (1 + np.sqrt(1/q))**2

# Define pdf
def f(lam):

    # Support of Marchenko--Pastur distribution
    if lam_minus <= lam <= lam_plus:

        return q/(2 * np.pi) * np.sqrt((lam_plus - lam) * (lam - lam_minus))/lam

    else:

        return 0
```

Marchenko–Pastur Example

```
# Get lam_vals
lam_vals = np.linspace(lam_minus, lam_plus, 100)

# Get density values
f_vals = [f(lam) for lam in lam_vals]

# Plot histogram
plt.hist(evals, density = True, bins = int(np.sqrt(N)), label = 'Simulated Distribution')

# Plot density
plt.plot(lam_vals, f_vals, label = 'Density')

# Add legend
plt.legend()

# Add x-label
plt.xlabel(r'$\lambda$')

# Add y-label
plt.ylabel(r'Density')

# Add title to plot
plt.title(r'Marchenko-Pastur Distribution')

# Save the figure
plt.savefig(path + r'ex4-4.png')

plt.show()

print(f'This script took {(time.perf_counter() - start_time)/60:.2f} minutes to run.')
```

Marchenko–Pastur Example Result



How to use it?

Consider eigenvalue λ of the covariance matrix.

- If $\lambda > \lambda_+$, then result consistent with signal.
- If $\lambda \leq \lambda_+$, the result is probably just random noise. Replace values less than λ_+ with mean of values less than λ_+ to make the covariance matrix more stable.

S&P 500 Constituent Example

Suppose S is the sample covariance matrix which we calculated from the S&P 500 constituent data before.

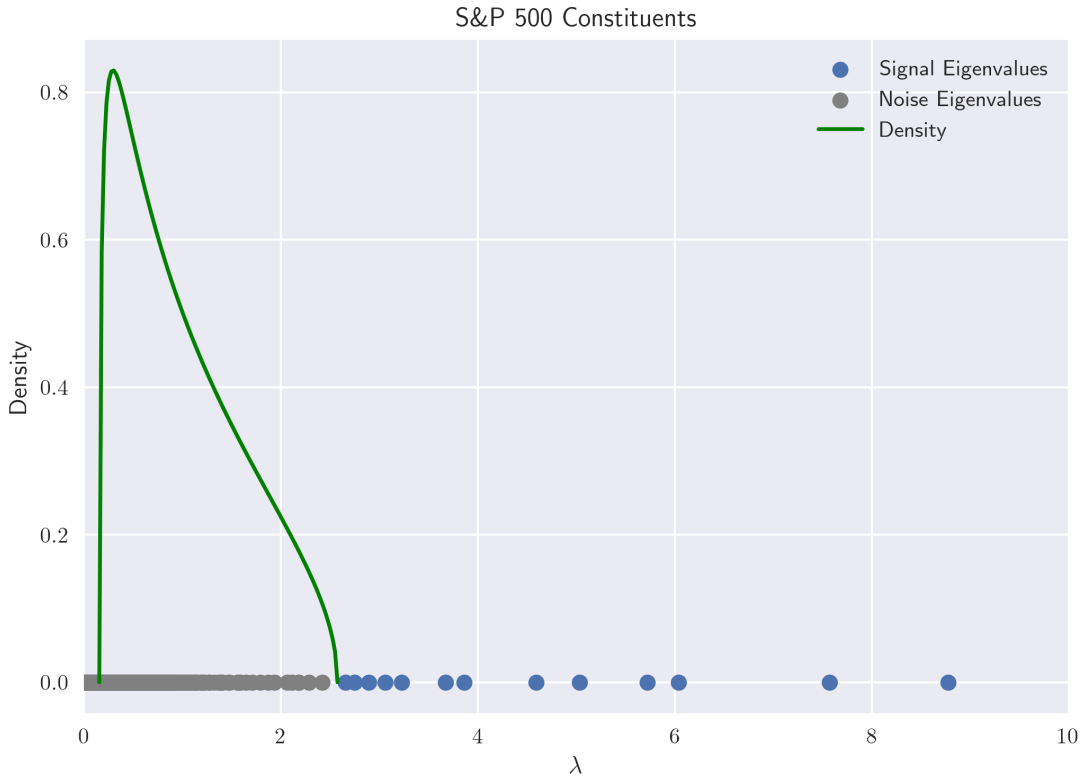
```
# Get the standard deviations
stds = np.sqrt(np.diag(S))

# Calculate correlation matrix
C = np.diag(1/stds) @ S @ np.diag(1/stds)

# Get the eigenvalues and vectors
evals, evecs =  np.linalg.eigh(C)

# Save q
q = data.shape[0]/data.shape[1]

# Get support of Marchenko-Pastur distribution
lam_minus, lam_plus = (1 - np.sqrt(1/q))**2, (1 + np.sqrt(1/q))**2

# Define pdf
def f(lam):

    # Support of Marchenko-Pastur distribution
    if lam_minus <= lam <= lam_plus:
        return q/(2 * np.pi) * np.sqrt((lam_plus - lam) * (lam - lam_minus))/lam
    else:
        return 0

# Get lam_vals
lam_vals = np.linspace(lam_minus, lam_plus, 100)

# Get density values
f_vals = f(lam_vals)
```

S&P 500 Constituent Example

```
# Plot histogram
plt.scatter(evals[evals > lam_plus], np.zeros(np.sum(evals > lam_plus)),
            label = 'Signal Eigenvalues')

plt.scatter(evals[evals <= lam_plus], np.zeros(np.sum(evals <= lam_plus)),
            label = 'Noise Eigenvalues', color = 'gray')

# Plot density
plt.plot(lam_vals, f_vals, label = 'Density', color = 'green')

# There are 3 eigenvalues much larger than 10
plt.xlim([0, 10])

# Add legend
plt.legend()

# Add x-label
plt.xlabel(r'$\lambda$')

# Add y-label
plt.ylabel(r'Density')

# Add title to plot
plt.title(r'S\&P 500 Constituents')

# Save the figure
plt.savefig(path + r'ex4-5.png')

plt.show()
```

S&P 500 Constituent Example

Only the values past λ_+ are consistent with signal. There are three eigenvalues greater than 10 which are not shown.

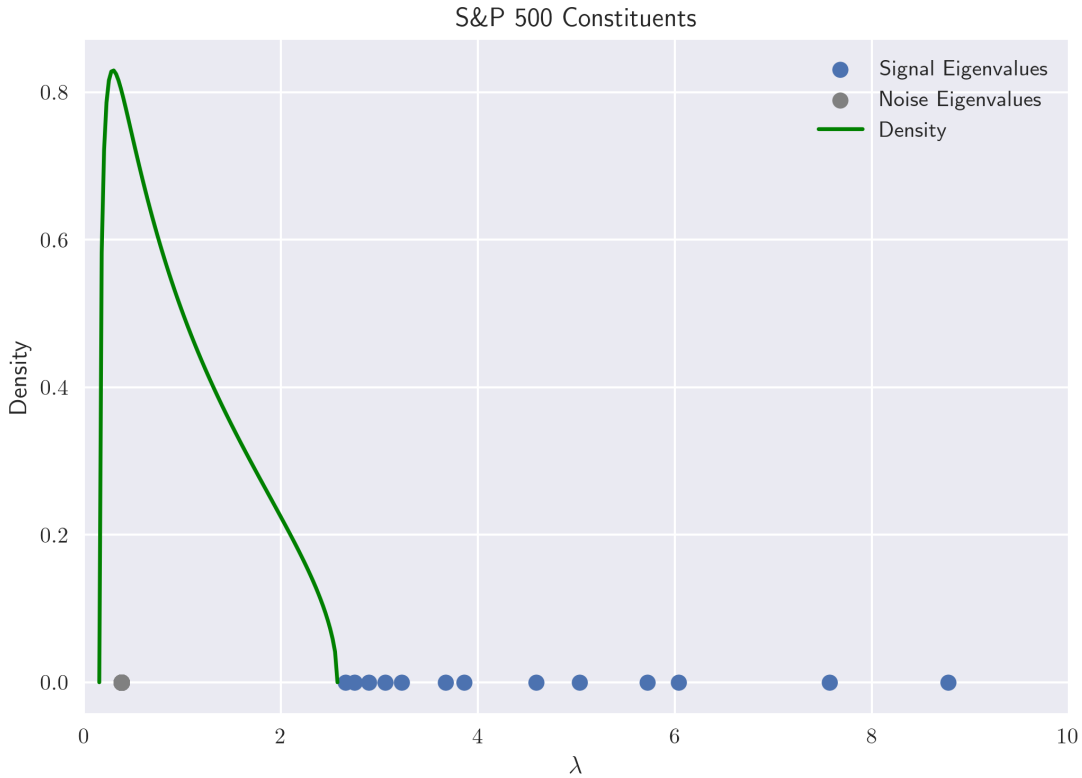


Clip Matrix

```
# Initialize new eigenvalues
evals_new = evals

# Replace noise eigenvalues with mean
evals_new[evals_new < lam_plus] = np.mean(evals_new[evals_new < lam_plus])

# Construct new correlation matrix
C_new = evecs @ np.diag(evals_new) @ evecs.T

# Make sure still correlation matrix
C_new = np.diag(1/np.sqrt(np.diag(C_new))) @ C_new @ np.diag(1/np.sqrt(np.diag(C_new)))

# Make new covariance matrix
S_new = np.diag(stds) @ C_new @ np.diag(stds)
```

S&P 500 Modified Correlation Matrix Eigenvalues

The signal eigenvalues are the same, while the noise eigenvalues have been replaced with the mean.



Stochastic Calculus

Source

I'm following these notes very closely:

`http://www.columbia.edu/~mh2078/FoundationsFE/IntroStochCalc.pdf`

Probability Triple

We assume we have the probability space $(\Omega, \mathcal{F}, P)$ where

- Ω is the universe of possible outcomes.
- $\mathcal{F}$ represents the σ-algebra of events in Ω.
- P is the "true" or physical probability measure.

Filtration

There is also a **filtration** $\{\mathcal{F}_t\}_{t\geq 0}$ of σ-algebras that models the evolution of information through time. Since information increases over time $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s < t$.

If it is know by time t whether or not an event E has occurred, then we have $E \in \mathcal{F}_t$. If we are working with a finite horizon $[0, T]$, then we can take $\mathcal{F} = \mathcal{F}_T$.

Stochastic Process

Definition

For a given probability space $(\Omega, \mathcal{F}, P)$, a **stochastic process** is a collection of random variables indexed by $\mathcal{T}$. We often write $\{X_t | t \in \mathcal{T}\}$ to denote a stochastic process, and we think of $\mathcal{T}$ as the time index. Sometimes we simply write X_t and only implicitly refer to the index.

Stochastic Process Example

Example

For a high yield bond portfolio, we can model the total number of defaulted bonds this year up to day t as a stochastic process. Denote the number of defaulted bonds on day t by N_t. In this case, $\mathcal{T} = \{1, 2, 3, \ldots, 252\}$, assuming there are 252 days when the market is open. Using our prior notation, the stochastic process is $\{N_t | t \in \mathcal{T}\}$.

$\mathcal{F}_t$-Adapted

Definition

We say that a stochastic process X_t is $\mathcal{F}_t$-**adapted** if for every t in $\mathcal{T}$ the information about X_t is contained in $\mathcal{F}_t$.

Brownian Motion

Definition

A stochastic process $\{W_t | t \geq 0\}$ is a **standard Brownian motion** if the following hold.

- BM.1 $W_0 = 0$.
- BM.2 *It has continuous sample paths.*
- BM.3 $W_{t_1} - W_{s_1}$ *and* $W_{t_2} - W_{s_2}$ *are independent for* $0 \leq s_1 \leq t_1 \leq s_2 \leq t_2$.
- BM.4 $W_t - W_s \sim \mathcal{N}(0, t - s)$ *for* $0 \leq s \leq t$.

Simulating Brownian Motion

Suppose that we want to simulate a Brownian motion on the interval $[0, T]$. Then construct a partition of the interval

$$0 = t_0 < t_1 < \ldots < t_{n-1} < t_n = T.$$

For $i = 1, 2, \ldots, n$, generate $Z_i \sim \mathcal{N}(0, 1^2)$. Then

$$\widetilde{W}_{t_k} = \begin{cases} 0, & k = 0 \\ \sum_{i=1}^{k} Z_i \sqrt{\Delta t_i}, & k = 1, 2, \ldots, n \end{cases}$$

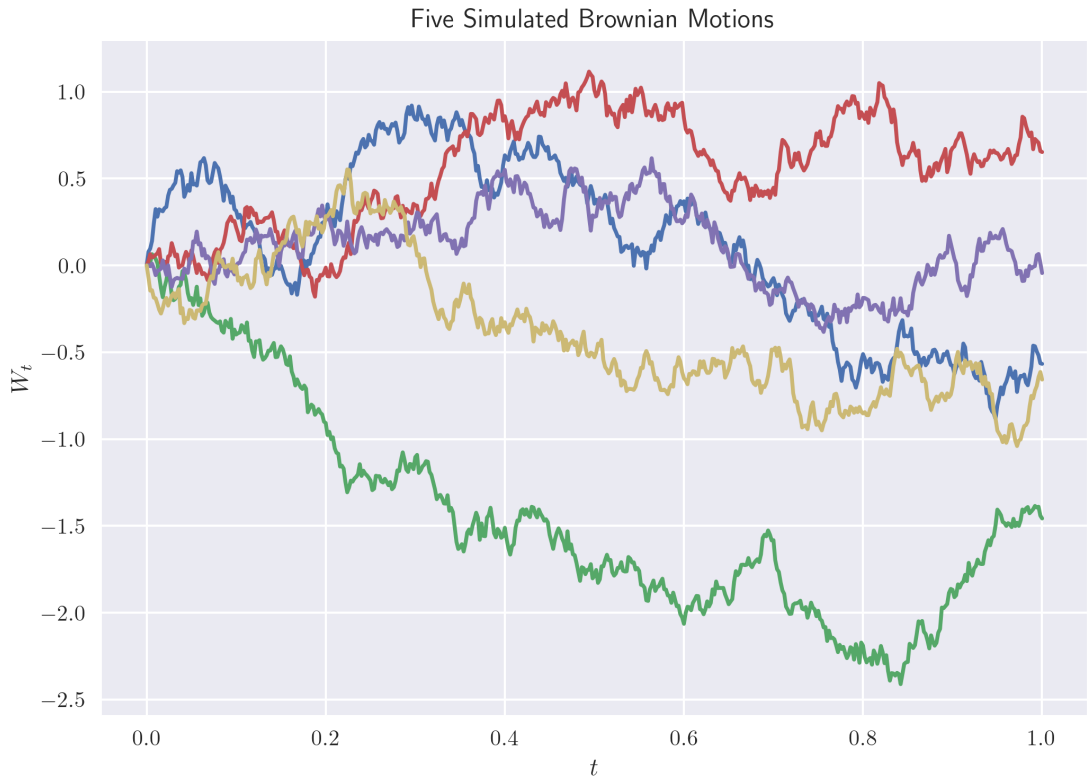
is "approximately" Brownian motion.

Python Code: Five Brownian Motions on [0, 1]

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'], plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set the random seed
np.random.seed(0)

# Break up so exact BM at n + 1 times
n = 500
# Simulate five Brownian motions
for _ in range(5):

    # Simulate Brownian motion for t in [0, 1]
    Z = norm.rvs(scale = np.sqrt(1/n), size = n)

    # Take the cumulative sum; add 0 for the t = 0 value
    W = np.insert(np.cumsum(Z), 0, 0)

    # Plot results
    plt.plot(np.linspace(0, 1, n + 1), W)

# Add x- and y-label as well as title
plt.xlabel(r'$t$'); plt.ylabel(r'$W_t$')
plt.title(r'Five Simulated Brownian Motions')

# Save the figure
plt.savefig(path + r'ex4-7.png')

plt.show()
```

Output

Five Brownian motions on the interval $[0, 1]$, using a uniform partition that breaks $[0, 1]$ into 500 subintervals.



Martingale

Definition

A stochastic process $\{X_t | 0 \leq t \leq \infty\}$ is a **martingale** with respect to the filtration $\mathcal{F}_t$ if the following hold.

M.1 $E[|X_t|] < \infty$ *for all* $t \geq 0$.

M.2 $E[X_{t+s}|\mathcal{F}_t] = X_t$ *for all* $t, s \geq 0$.

Martingale Example

Example

Prove the following are martingales.

(a) W_t

(b) $W_t^2 - t$

(c) $\exp\left(\theta W_t - \frac{\theta^2 t}{2}\right)$

Quadratic Variation

Definition

Let X_t be some stochastic process. The **quadratic variation** of a stochastic process X_t is

$$\lim_{\|\mathcal{P}\| \to 0} \sum_{k=1}^{n} \left(X_{t_k} - X_{t_{k-1}} \right)^2,$$

where $\mathcal{P}$ is an arbitrary partition of $[0, T]$ and $\|\mathcal{P}\| = \max_k \{\Delta t_k\}$ is the mesh of the partition.

Quadratic Variation Example

Example

Compute the quadratic variation of the deterministic process $X_t = t^2$.

Quadratic Variation Differentiable Function

Any differentiable function will end up having quadratic variance 0, like we saw for t^2 in our example.

Quadratic Variation of Brownian Motion

Theorem

The quadratic variation of a standard Brownian motion $\{W_t | 0 \leq t \leq T\}$ *is equal to* T *with probability 1.*

Theorem (Levey's Theorem)

A continuous martingale is a standard Brownian motion if and only if its quadratic variation over each interval $[0, T]$ *is equal to* T*.*

Approximate Quadratic Variation

We will approximate Brownian motion in the interval $[0, 1]$ with finer and finer partitions. We will see that as the partitions get finer, the quadratic variation approaches a straight line.

Approximate Quadratic Variation Python Code

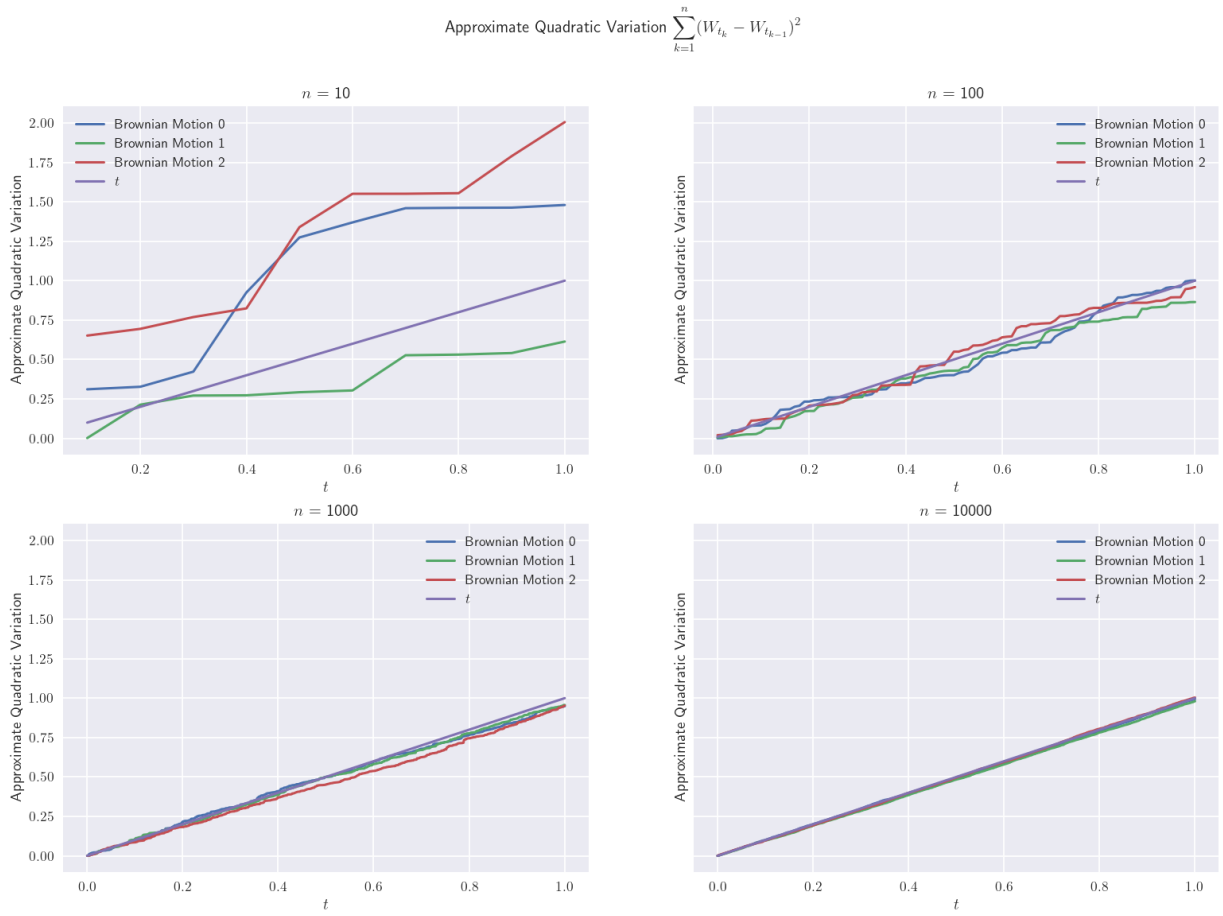
```
# Import modules
import numpy as np, matplotlib.pyplot as plt
from scipy.stats import norm

# Use LaTeX and increase resolution
plt.rcParams['text.usetex'],  plt.rcParams['figure.dpi'] = True, 300

# Use Seaborn style
plt.style.use('seaborn-v0_8')

# Set the random seed
np.random.seed(0)

# Define n-values
n_vals = [10, 100, 1000, 10000]

# Set up subplots
fig, ax = plt.subplots(2, 2, sharey = True, figsize = (15, 10), dpi = 125)

# Simulate five Brownian motions
for i, n in enumerate(n_vals):

    # Get row and column
    row, col = i//2, i % 2

    # Difference of two Brownian motions is normal
    dW = norm.rvs(scale = np.sqrt(1/n), size = (3, n))

    # Calculate quadratic variance
    quad_var = np.cumsum(dW**2, axis = 1)
```

Approximate Quadratic Variation Python Code

```
        # Plot results
        for j in range(3):
            ax[row, col].plot(np.linspace(1/n, 1, n), quad_var[j, :], label = f'Brownian
          Motion {j}')
        # Plot t
        ax[row, col].plot(np.linspace(1/n, 1, n), np.linspace(1/n, 1, n), label = r'$t$')
        # Add x-label
        ax[row, col].set_xlabel(r'$t$')
        # Add y-label
        ax[row, col].set_ylabel(r'Approximate Quadratic Variation')
        # Give plot a title
        ax[row, col].set_title(f'$n$ = {n}')
        # Add legend
        ax[row, col].legend()
# Add title to plot
plt.suptitle(r'Approximate Quadratic Variation $\displaystyle\sum_{k = 1}^{n} (W_{t_k} -
     W_{t_{k - 1}})^2$')
# Save the figure
plt.savefig(path + r'ex4-8.png')
plt.show()
```

Approximate Quadratic Variation Result

From the four subplots, we see that $\lim_{n\to\infty} \sum_{k=1}^{n} \left(W_{t_k} - W_{t_{k-1}}\right)^2 = T$.



Itô Integrals

Definition

The **Itô Integral** of X_t with respect to standard Brownian motion

$$\int_0^T X_t \, dW_t = \lim_{\|P\| \to 0} \sum_{k=1}^{n} X_{t_{k-1}} (W_{t_k} - W_{t_{k-1}}),$$

where $\mathcal{P} = (t_0, t_1, \ldots, t_n)$ is an arbitrary partition of $[0, T]$.

Itô Integral Example

Example

Assuming the Itô integral exists, compute $\int_0^T W_t \, dW_t$.

Solution.

$$\begin{aligned}
\int_0^T W_t \, dW_t &= \lim_{n\to\infty} \sum_{k=1}^{n} W_{t_{k-1}} \left(W_{t_k} - W_{t_{k-1}}\right) \\
&= \lim_{n\to\infty} \frac{1}{2} \sum_{k=1}^{n} \left[W_{t_k}^2 - W_{t_{k-1}}^2 - (W_{t_k} - W_{t_{k-1}})^2\right] \\
&= \frac{1}{2}(W_T^2 - W_0^2) - \frac{1}{2} \lim_{n\to\infty} \sum_{k=1}^{n} (W_{t_k} - W_{t_{k-1}})^2 \\
&= \frac{1}{2}(W_T^2 - W_0^2) - \frac{1}{2} T \\
&= \frac{1}{2} W_T^2 - \frac{1}{2} T.
\end{aligned}$$

Itô Integral Python Example

Example

Check the previous result by running a simulation in Python.

Itô Integral Python Example

```
# Import modules
import numpy as np
from scipy.stats import norm

# Set the random seed
np.random.seed(0)

# Set n, so there are n + 1 places where BM exact
n = 1_000_000

# Define T
T = 1

# Simulate independent increment
dW = norm.rvs(scale = np.sqrt(T/n), size = n)

# Sum up to get BM
W = np.insert(np.cumsum(dW), 0, 0)

# Approximate Ito integral; using left endpoints
integral_simulated = np.sum(W[0:-1] * dW)

# Write analytic result
integral_analytic = 1/2 * W[-1]**2 - T/2

print(f'The simulated result is {integral_simulated:.3f} while the analytic result is {
    integral_analytic:.3f}.')
```

The output reads:

The simulated result is 0.643 while the analytic result is 0.643.

Law of Iterated Expectations

Suppose that $s \leq t \leq T$. Then

$$E[X_T|\mathcal{F}_s] = E\Big[E[X_T|\mathcal{F}_t]\Big|\mathcal{F}_s\Big].$$

Instead of writing $E[X_T|\mathcal{F}_t]$ we often write $E_t[X_T]$. Using this notation, the Law of Iterated Expectations is

$$E_s[X_T] = E_s\Big[E_t[X_T]\Big].$$

Itô Isometry

Theorem (Itô Isometry)

We have

$$E\left[\left(\int_0^T X_t \, dW_t\right)^2\right] = E\left[\int_0^T X_t^2 \, dt\right]$$

whenever

$$E\left[\int_0^T X_t^2 \, dt\right] < \infty.$$

Itô Integral Example

Example

Prove that the stochastic process

$$\int_0^T X_t \, dW_t$$

is a martingale.

Stochastic Differential Equation

Definition

We define the **stochastic differential equation**

$$dX_t = a(t, X_t)\ dt + b(t, X_t)\ dW_t$$

to mean

$$X_t = X_0 + \int_0^t a(s, X_s)\ ds + \int_0^t b(s, X_s)\ dW_s.$$

Box Calculus

These rules hold for stochastic differential equations

$$dt^2 = 0, \quad dt\ dW_t = 0, \quad \text{and} \quad dW_t^2 = dt$$

Box Calculus Example

Example

Suppose

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t.$$

Use box Calculus to compute dS_t^2.

Itô's Lemma

Theorem

Suppose that $dX_t = \mu_t\ dt + \sigma_t\ dW_t$, *and* $Z_t = f(t, X_t)$, *where* $f(t, x)$ *is continuously differentiable at least once with respect to* t *and at least twice with respect to* x. *Then*

$$dZ_t = \left(\frac{\partial f}{\partial t}(t, X_t) + \mu_t \frac{\partial f}{\partial x}(t, X_t) + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) \right) dt + \sigma_t \frac{\partial f}{\partial x}(t, X_t)\ dW_t.$$

Itô's Lemma Example

Example

Suppose $dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$. Use Itô's Lemma to find $d(\log S_t)$.